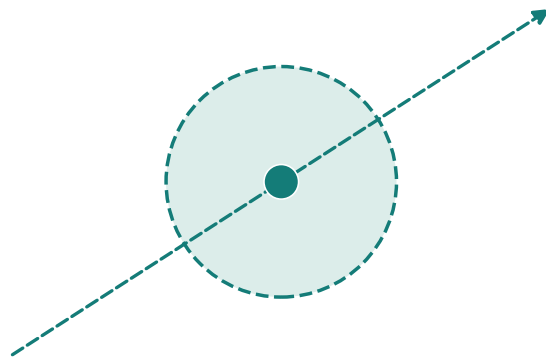


$$\ell + r = 0, \quad \ell^2 = r^2 = 0 \quad \implies \quad \mathbf{R}_B \simeq \operatorname{Spec} \mathbb{C}[\ell]/(\ell^2)$$

Tangent direction



One geometric point

Halo = nilpotent structure (schematic)

$$T_p \mathbf{R}_B = \mathbb{C}(1, -1)$$

$$\ell_t = tx + t^2 u, \quad r_t = -tx + t^2 v$$



$$\ell_t^2 = t^2 x^2 \pmod{t^3}$$
$$x \neq 0 \implies \text{obstructed}$$

A nonzero Zariski tangent vector need not integrate to a local family.